Energy Measurement

Motor Meters

- Moving system revolves continuously.
- Speed is proportional to power.
- Total number of revolutions in a given interval of time is proportional to energy.
- Meter constant is the number of revolutions made per KWH.

Theory of Induction Type Energy Meter

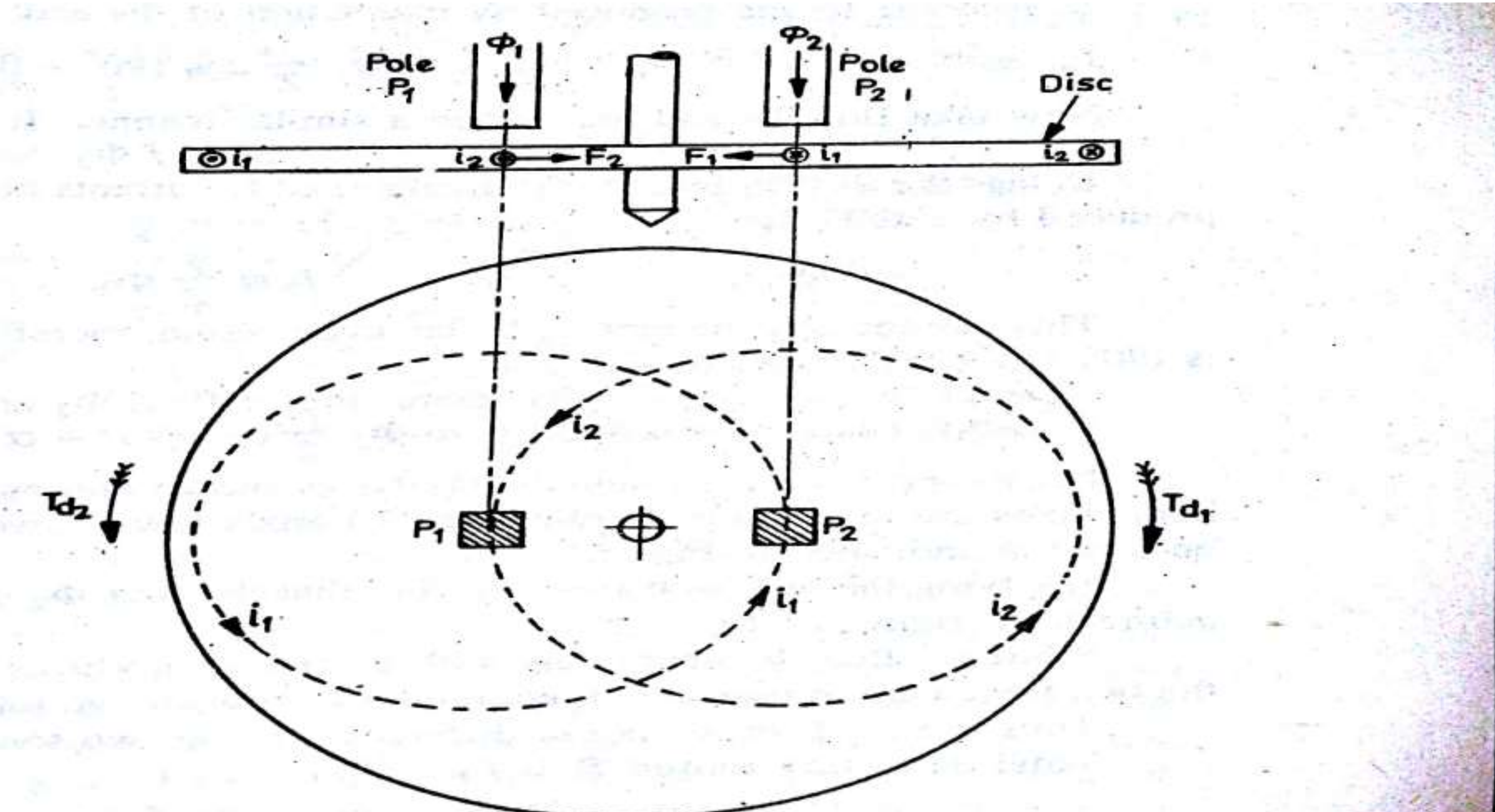


Fig. 5.1. Principle of working of an induction type instrument.

→ Two fluxes

→ Produce emfs in a metallic disc

→ Pole $\left.\begin{array}{l} P_1 \rightarrow \phi_1 \rightarrow i_1 \\ P_2 \rightarrow \phi_2 \rightarrow i_2 \end{array}\right]$ eddy currents

→ Two torques are produced by $\phi_1 i_2$ & $\phi_2 i_1$.

Let ϕ_1 & ϕ_2 be the instantaneous values of two fluxes having a phase difference of β.

$$\phi_1 = \phi_{m1} \sin \omega t \quad \phi_2 = \phi_{m2} \sin(\omega t - \beta)$$

$$(\phi_1)_r = \phi_{m1}/\sqrt{2} \quad (\phi_2)_r = \phi_{m2}/\sqrt{2}$$

Emf due to ϕ_1 $\quad e_1 \propto \left[-\frac{d\phi_1}{dt}\right] \propto \left[-f\,\phi_{m1} \cos \omega t\right]$

e_1 lags ϕ_1 by 90°

Rms value of e_1 (E_1) α $f(\phi_1)_r$

Rms value of eddy current $(I_1) \propto \frac{f}{Z}(\phi_1)_r$ [Z → eddy current path impedance]

I_1 lags E_1 by angle 'α'

Phase angle between ϕ_2 & $I_1 = 90° - \beta + \alpha$

Average torque produced by interaction of $(\phi_2)_r$ & I_1

$$T_{d1} \propto (\phi_2)_r I_1 \cos(90° - \beta + \alpha) \propto (\phi_2)_r (\phi_1)_r \frac{f}{Z} \cos(90° - \beta + \alpha)$$

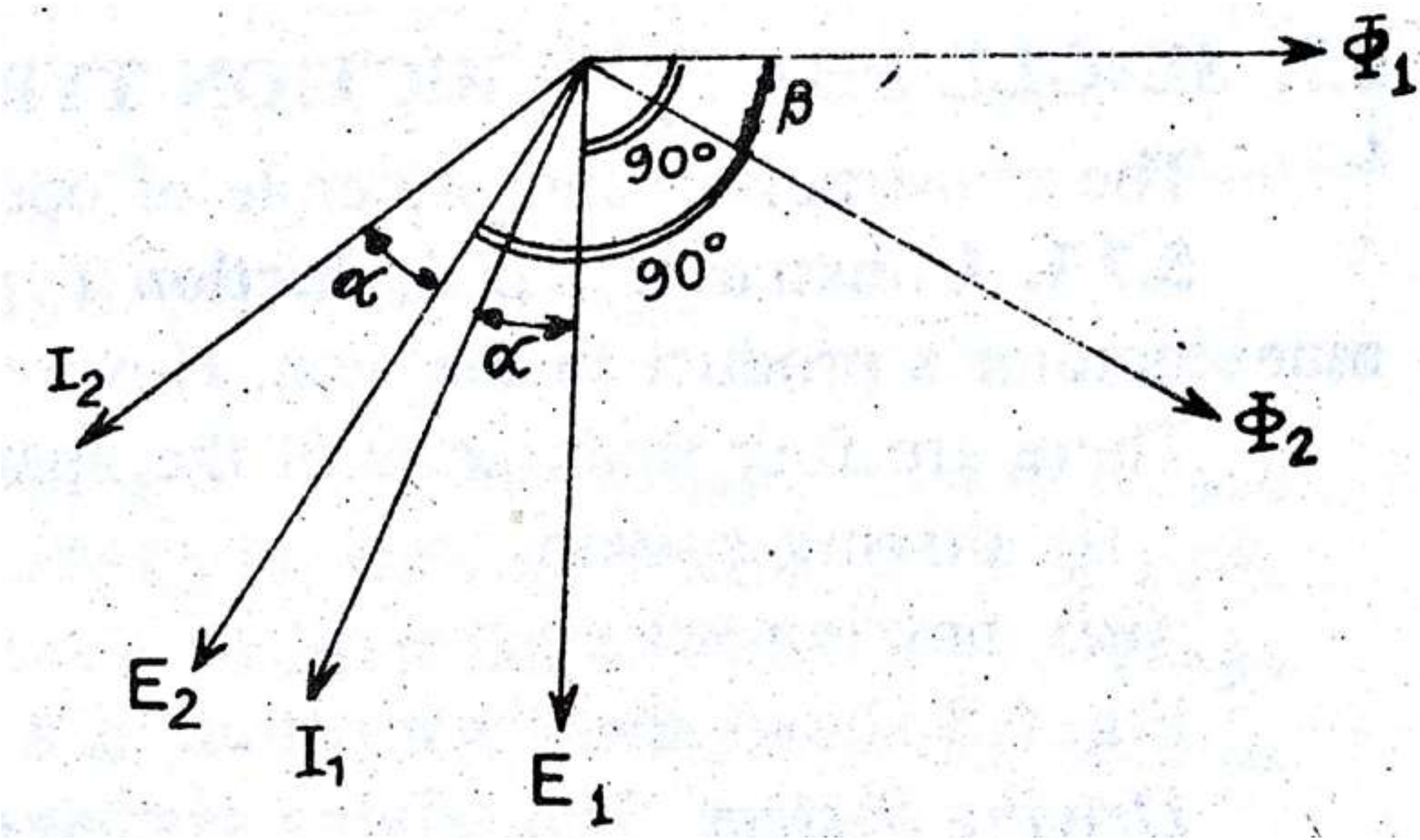
Similarly, average torque produced by interaction of $(\phi_1)_r$ & I_2

$$T_{d2} \propto (\phi_1)_r (\phi_2)_r \frac{f}{Z} \cos(90° + \alpha + \beta)$$ [∵ angle bet. ϕ_1 & I_2 is $(90° + \alpha + \beta)$]

$T_d = T_{d1} - T_{d2}$ (Torques act in opposite directions)

$$\Rightarrow T_d \propto (\phi_1)_r (\phi_2)_r \frac{f}{Z} \sin\beta \cos\alpha$$

Phasor Diagram of Induction Type Instrument



Single Phase Induction Type Meters

- Four main parts:

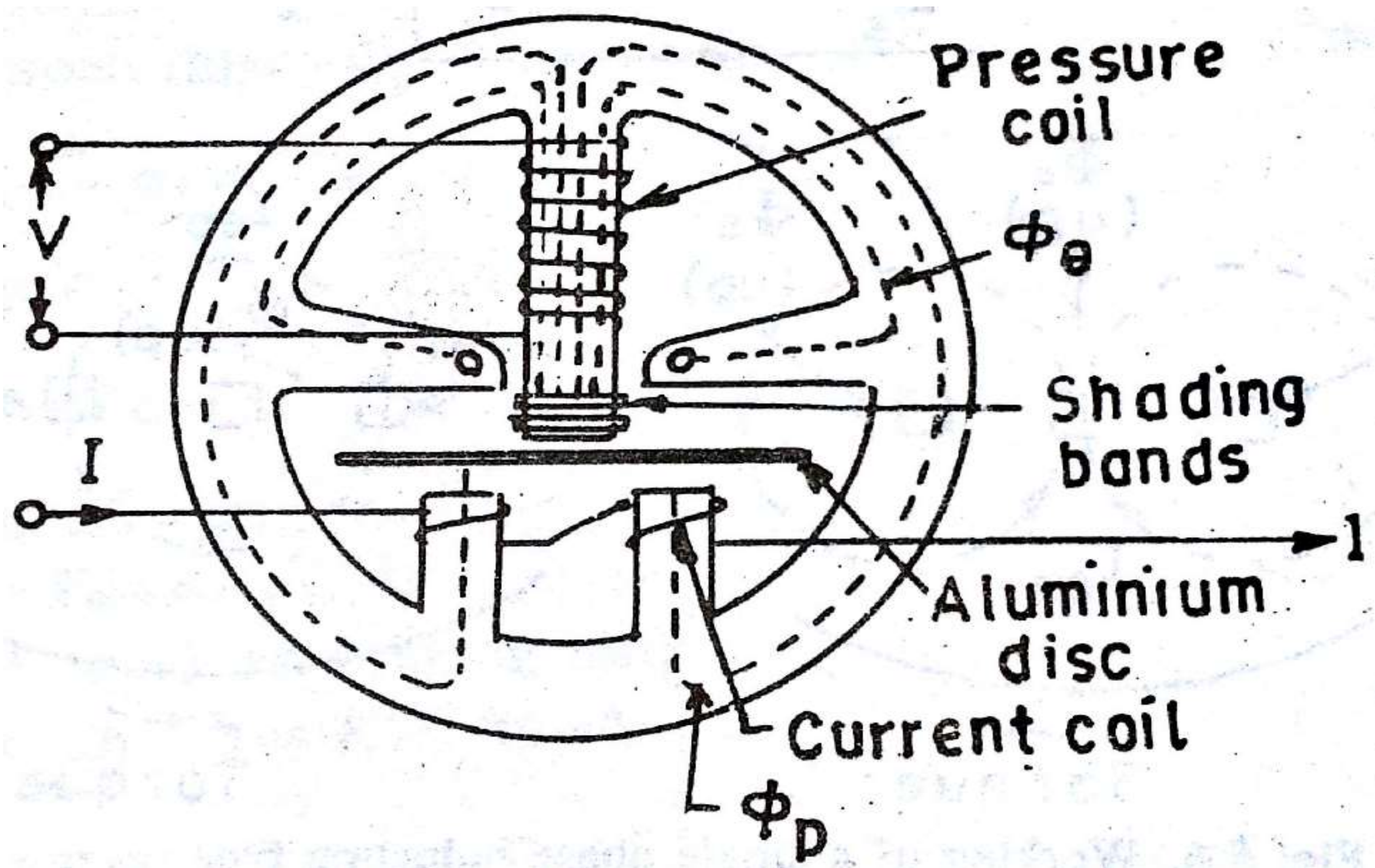
  * Driving System (Two electromagnets- PC & CC)

  * Moving System (Aluminium Disc)

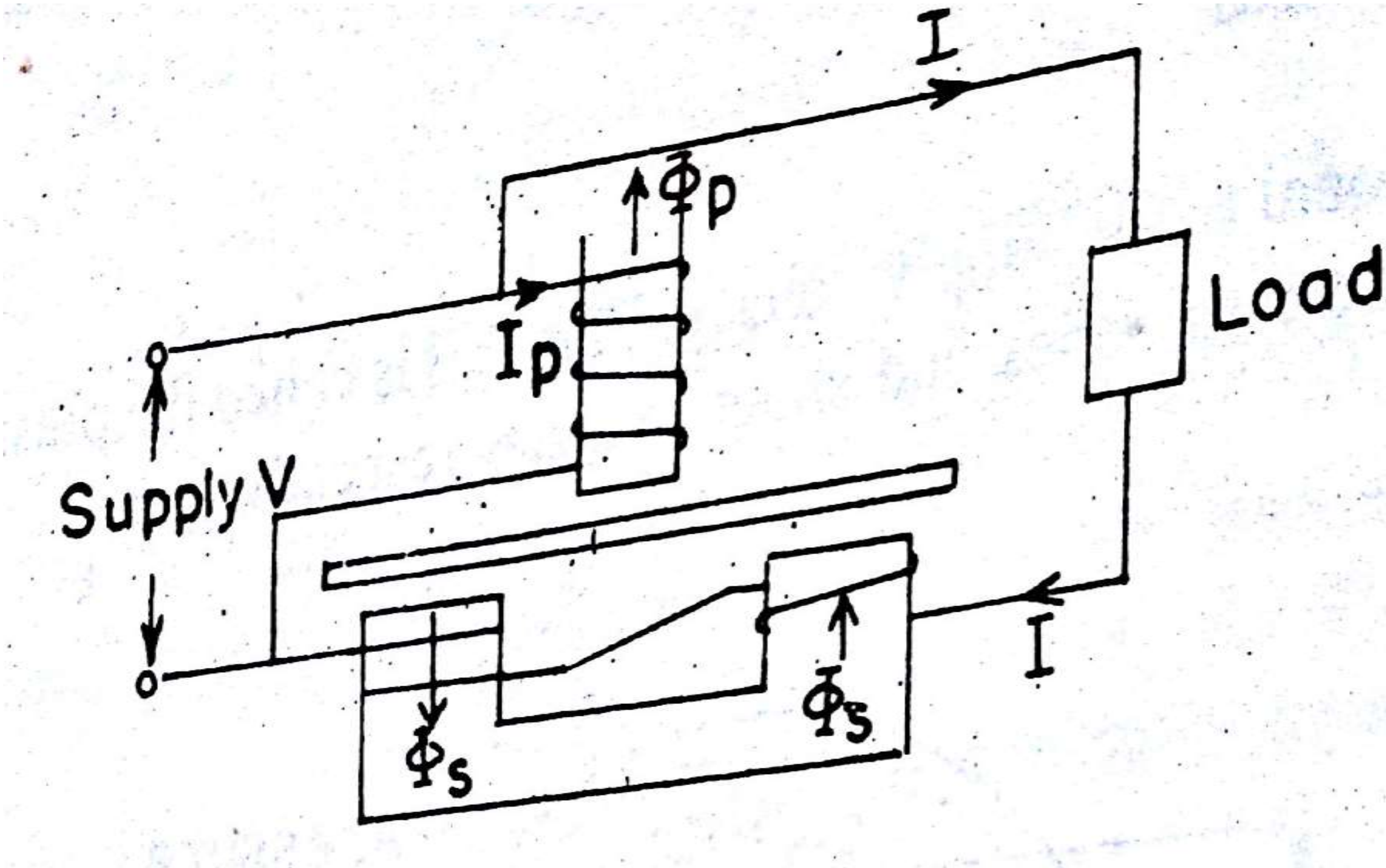
  * Braking System (Permanent Magnet)

  *Registering System (Counting mechanism with gears)

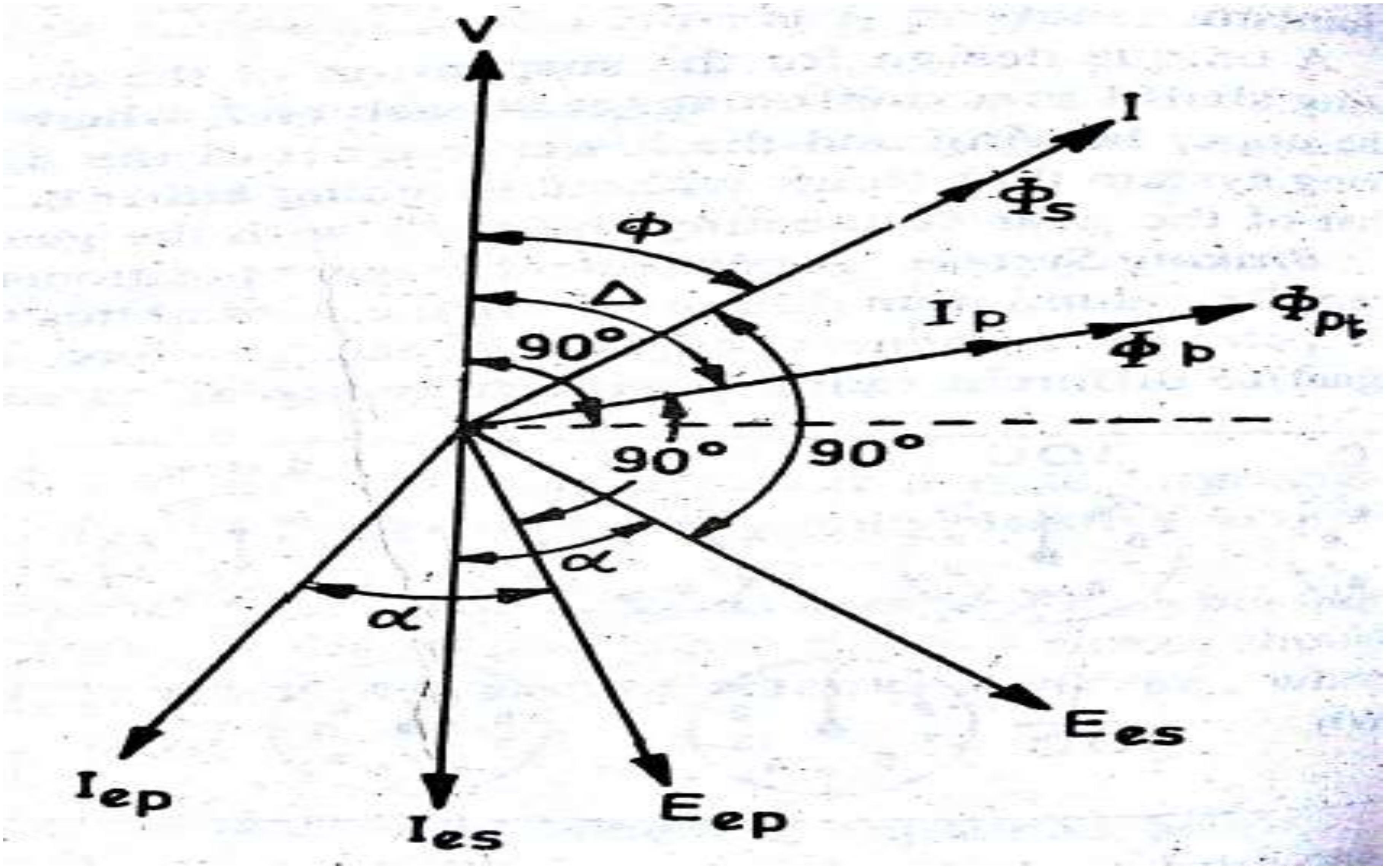
Single Phase Energy Meter



Working



Phasor Diagram



Driving Torque

$$T_d \propto \Phi_1 \Phi_2 \frac{f}{Z} \sin\beta \cos\alpha$$

$$\Rightarrow T_d = K_1 \Phi_1 \Phi_2 \frac{f}{Z} \sin\beta \cos\alpha$$

For given energy meter

$$T_d = K_1 \Phi_p \Phi_s \frac{f}{Z} \sin(\Delta - \Phi) \cos\alpha$$

But $\Phi_p \propto V$ & $\Phi_s \propto I$

$$\therefore T_d = K_2 V I \frac{f}{Z} \sin(\Delta - \Phi) \cos\alpha$$

If f, Z and α are constants,

$$T_d = K_3 VI \sin(\Delta - \Phi)$$

For a steady speed N, braking torque $(T_B) = K_4 N$

At steady speed

$$T_d = T_B \quad \Rightarrow K_4 N = K_3 VI \sin(\Delta - \Phi)$$

or $N = K VI \sin(\Delta - \Phi)$

For $\Delta = 90^\circ$, $N = K VI \cos\Phi$

Total Number of revolutions $= \int N\,dt$

$$= K \int VI \cos\Phi \, dt$$

$$= K \text{ (energy)}$$

Braking

- Speed of moving system is controlled by braking.
- A permanent magnet induces eddy current in some part of the moving system.
- Braking torque is proportional to the speed of the moving system.
- The moving system is usually an aluminium disc.

Emf generated in the disc $(e) = K_1 \phi n$

where $\phi \rightarrow$ flux of permanent magnet

$n \rightarrow$ speed of rotation

Eddy current produced $(i) = \frac{e}{r} = \frac{K_1 \phi n}{r}$

where $r \rightarrow$ resistance of eddy current path

Braking torque is produced by interaction of eddy current and the field of the permanent magnet.

Braking torque $(T_B) = K_2 \phi i R = K_1 K_2 \phi^2 n R/r = K_3 \phi^2 n R/r$

where 'R' $\rightarrow$ effective radius of disc from axis

$$T_B = K_4 \phi^2 n/r \qquad \text{(For constant R)}$$

Moving system attains a steady speed when the driving torque = braking torque.

$$T_B = K_5 N \qquad \text{(At steady speed N)}$$

$$T_B = T_d$$

$$\Rightarrow N = K_6 T_d \qquad \text{where } K_6 = \frac{r}{K_3 \phi^2 R}$$

$$\Rightarrow N \propto r$$

$$\propto \frac{1}{\phi^2}$$

$$\propto \frac{1}{R}$$